WAVE OPTICS

SUBJECTIVE ASSIGNMENT - 1

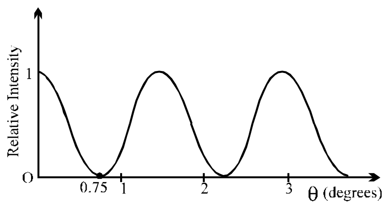
1. In a Young's double slit experiment for interference of light, the slits are 0.2 cm apart and are illuminated by yellow light ($\lambda = 600$ nm). What would be the fringe width on a screen placed 1 m from the plane of slits if the whole system is immersed in water of index 4/3?

WO0001

2. In Young's double slit experiment the slits are 0.5 mm apart and the interference is observed on a screen at a distance of 100 cm from the slit. It is found that the 9th bright fringe is at a distance of 7.5mm from the second dark fringe from the centre of the fringe pattern on same side. Find the wavelength of the light used.

WO0002

3. Light of wavelength 520 nm passing through a double slit, produces interference pattern of relative intensity versus angular position θ as shown in the figure. Find the separation d between the slits.



WO0003

4. In a Young's double slit experiment, two wavelengths of 500 nm and 700 nm were used. What is the minimum distance from the central maximum where their maximas coincide again? (Take $D/d = 10^3$. Symbols have their usual meanings.) **[IIT-JEE 2004]**

WO0004

5. In a YDSE apparatus, d = 1mm, λ = 600 nm and D = 1m. The slits individually produce same intensity on the screen. Find the minimum distance between two points on the screen having 75% intensity of the maximum intensity.

WO0005

6. The distance between two slits in a YDSE apparatus is 3mm. The distance of the screen from the slits is 1m. Microwaves of wavelength 1 mm are incident on the plane of the slits normally. Find the distance of the first maxima on the screen from the central maxima. Also find the total number of maxima on the screen.

WO0006

7. One slit of a double slit experiment is covered by a thin glass plate of refractive index 1.4 and the other by a thin glass plate of refractive index 1.7. The point on the screen, where central bright fringe was formed before the introduction of the glass sheets, is now occupied by the 5^{th} bright fringe. Assuming that both the glass plates have same thickness and wavelength of light used is 4800 Å, find their thickness.

WO0007

8. A monochromatic light of $\lambda = 5000$ Å is incident on two slits separated by a distance of 5×10^{-4} m. The interference pattern is seen on a screen placed at a distance of 1 m from the slits. A thin glass plate of thickness 1.5×10^{-6} m & refractive index $\mu = 1.5$ is placed between one of the slits & the screen. Find the intensity at the centre of the screen, if the intensity there is I_0 in the absence of the plate. Also find the lateral shift of the central maximum.

WO0008

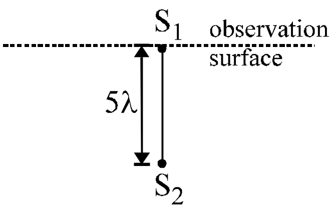
9. In a biprism experiment with sodium light, bands of width 0.0195 cm are observed on screen at 100cm from slit. On introducing a convex lens 30 cm away from the slit between biprism and screen, two images of the slit are seen 0.7 cm apart on screen. Calculate the wavelength of sodium light.

WO0009

10. A long narrow horizontal slit lies 1mm above a plane mirror as in Lloyd's mirror. The interference pattern produced by the slit and its image is viewed on a screen distant 1m from the slit. The wavelength of light is 600nm. Find the distance of first maximum above the mirror.

WO0010

11. Two microwave coherent point sources emitting waves of wavelength λ are placed at 5λ distance apart. The interference is being observed on a flat non-reflecting surface along a line passing through one source, in a direction perpendicular to the line joining the two sources (refer figure). Considering λ as 4 mm, calculate the positions of maxima and draw shape of interference pattern. Take initial phase difference between the two sources to be zero.



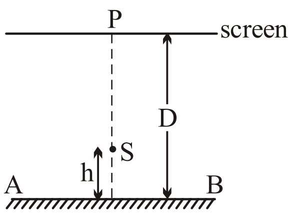
WO0011

12. A point source S emitting light of wavelength 600 nm is placed at a very small height h above the flat reflecting surface AB (see figure). The intensity of the reflected light is 36% of the incident intensity. Interference fringes are observed on a screen placed parallel to the reflecting surface at a very large distance D from it. **[IIT-JEE 2002]**

(i) What is the shape of the interference fringes on the screen?

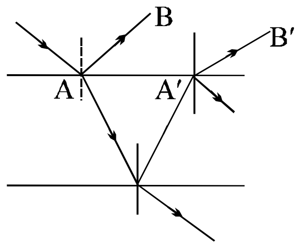
(ii) Calculate the ratio of the minimum to the maximum intensities in the interference fringes formed near the point P (shown in the figure).

(iii) If the intensities at point P corresponds to a maximum, calculate the minimum distance through which the reflecting surface AB should be shifted so that the intensity at P again becomes max.



WO0012

13. A ray of light of intensity I is incident on a parallel glass-slab at a point A as shown in figure. It undergoes partial reflection and refraction. At each reflection 20% of incident energy is reflected. The rays AB and A′ B′ undergo interference. Find the ratio I_{max}/I_{min}. [Neglect the absorption of light]



WO0013

WAVE OPTICS

SUBJECTIVE ASSIGNMENT - II

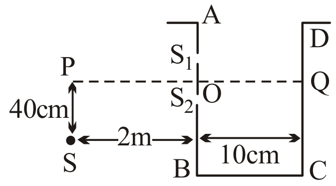
1. A thin glass plate of thickness t and refractive index μ is inserted between screen & one of the slits in a Young's experiment. If the intensity at the centre of the screen is I, what was the intensity at the same point prior to the introduction of the sheet.

WO0014

2. In Young's experiment, the source is red light of wavelength 7×10^{-7} m . When a thin glass plate of refractive index 1.5 at this wavelength is put in the path of one of the interfering beams, the central bright fringe shifts by 10^{-3} m to the position which was previously occupied by the 5^{th} bright fringe. Find the thickness of the plate. When the source is now changed to green light of wavelength 5×10^{-7} m, the central fringe shifts to a position initially occupied by the 6^{th} bright fringe due to red light without the plate. Find the refractive index of glass for the green light. Also estimate the change in fringe width due to the change in wavelength. **[IIT-JEE 1997 C]**

WO0015

3. A vessel $ABCD$ of 10 cm width has two small slits S_1 and S_2 sealed with identical glass plates of equal thickness. The distance between the slits is 0.8 mm. POQ is the line perpendicular to the plane AB and passing through O, the middle point of S_1 and S_2. A monochromatic light source is kept at S, 40cm below P and 2 m from the vessel, to illuminate the slits as shown in the figure below. Calculate the position of the central bright fringe on the other wall CD with respect to the line OQ. Now, a liquid is poured into the vessel and filled up to OQ. The central bright fringe is found to be at Q. Calculate the refractive index of the liquid. **[IIT-JEE 2001]**



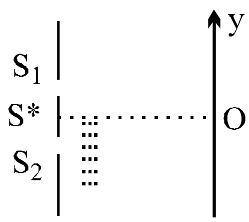
WO0016

4. The Young's double slit experiment is done in a medium of refractive index 4/3. A light of 600 nm wavelength is falling on the slits having 0.45 mm separation. The lower slit S_2 is covered by a thin glass sheet of thickness 10.4 μm and refractive index 1.5. The interference pattern is observed on a screen placed 1.5 m from the slits as shown **[IIT-JEE'99]**

(i) Find the location of the central maximum (bright fringe with zero path difference) on the y-axis.

(ii) Find the light intensity at point O relative to the maximum fringe intensity.

(iii) Now, if 600 nm light is replaced by white light of range 400 to 700 nm, find the wavelengths of the light that form maxima exactly at point O. [All wavelengths in this problem are for the given medium of refractive index 4/3. Ignore dispersion]

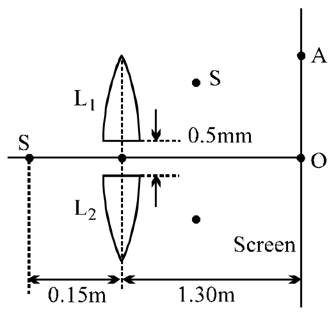


WO0017

5. In a Young's experiment, the upper slit is covered by a thin glass plate of refractive index 1.4 while the lower slit is covered by another glass plate having the same thickness as the first one but having refractive index 1.7. Interference pattern is observed using light of wavelength 5400 Å. It is found that the point P on the screen where the central maximum ($n = 0$) fell before the glass plates were inserted now has 3/4 the original intensity. It is further observed that what used to be the 5th maximum earlier, lies below the point P while the 6th minimum lies above P. Calculate the thickness of the glass plate. (Absorption of light by glass plate may be neglected). **[IIT-JEE 1997]**

WO0018

6. In the figure shown S is a monochromatic point source emitting light of wavelength λ = 500 nm. A thin lens of circular shape and focal length 0.10 m is cut into two identical halves L_1 and L_2 by a plane passing through a diameter. The two halves are placed symmetrically about the central axis SO with a gap of 0.5 mm. The distance along the axis from S to L_1 and L_2 is 0.15 m, while that from L_1 & L_2 to O is 1.30 m. The screen at O is normal to SO. **[IIT-JEE 1993]**

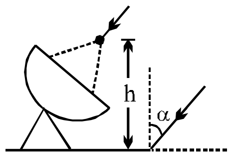


(i) If the third intensity maximum occurs at the point A on the screen, find the distance OA.

(ii) If the gap between L_1 & L_2 is reduced from its original value of 0.5 mm, will the distance OA increase, decrease or remain the same ?

WO0019

7. Radio waves coming at angle α to vertical are recieved by a radar after reflection from a nearby water surface & directly. What should be height of antenna from water surface so that it records a maximum intensity. (wavelength = λ).

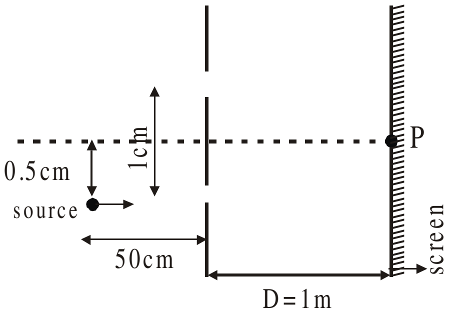


WO0020

8. One radio transmitter A operating at 60.0 MHz is 10.0 m from another similar transmitter B that is 180° out of phase with transmitter A. How far must an observer move from transmitter A toward transmitter B along the line connecting A and B to reach the nearest point where the two beams are in phase?

WO0021

9. In a typical Young's double slit experiment a point source of monochromatic light is kept as shown in the figure. If the source is given an instantaneous velocity v = 1 mm per second towards the screen, then the instantaneous velocity of central maxima is given as $\alpha \times 10^{-\beta}$ m/s upward. Find the value of $\alpha+\beta$.

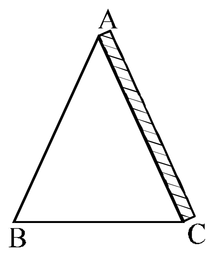


WO0022

10. A prism ($\mu_P = \sqrt{3}$) has an angle of prism $A = 30°$. A thin film ($\mu_f = 2.2$) is coated on face AC as shown in the figure. Light of wavelength 550 nm is incident on the face AB at 60° angle of incidence. Find **[IIT-JEE 2003]**

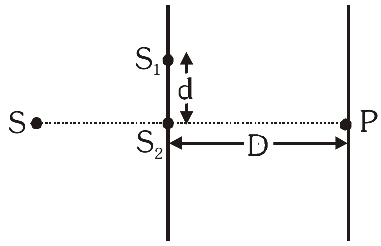
(i) the angle of its emergence from the face AC and

(ii) the minimum thickness (in nm) of the film for which the emerging light is of maximum possible intensity.



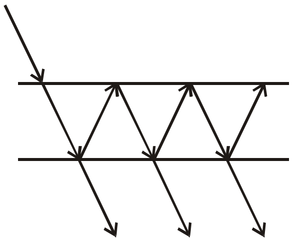
WO0023

11. In a YDSE experiment two slits S_1 and S_2 have separation of $d = 2$ mm. The distance of the screen is $D = 8/5$ m. Source S starts moving from a very large distance towards S_2 perpendicular to S_1S_2 as shown in figure. The wavelength of monochromatic light is 500 nm. The number of maximas observed on the screen at point P as the source moves towards S_2 is 3995 + n. Find the value of n.



WO0024

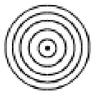
12. A narrow beam of light has entered a large thin glass plate. Each refraction is accompanied by reflection of one third of the beam's energy. What percentage of the light energy is transmitted through the plate?



WO0025

ANSWER KEY

SUBJECTIVE ASSIGNMENT - 1

1. Ans. 0.225 mm
2. Ans. 5000Å
3. Ans. 1.99×10^{-2} mm
4. Ans. 3.5 mm
5. Ans. 0.2 mm
6. Ans. 35.35 cm approximate, 5
7. Ans. 8 μm
8. Ans. 0, 1.5 mm
9. Ans. $\lambda = 5850$ Å
10. Ans. 0.15 mm
11. Ans. 48, 21, $\frac{32}{3}$, $\frac{9}{2}$, 0 m.m.; 
12. Ans. (i) circular, (ii) $\frac{1}{16}$, (C) 3000Å
13. Ans. 81 : 1

SUBJECTIVE ASSIGNMENT - II

1. Ans. $I_0 = I\sec^2\left[\frac{\pi(\mu-1)t}{\lambda}\right]$
2. Ans. 7 μm, 1.6, $\frac{400}{7}$ μm (decrease)
3. Ans. (i) y = 2 cm, (ii) m = 1.0016
4. Ans. (i) y = – 13/3 mm, (ii) intensity at O = $0.75I_{max}$ (iii) 650 nm, 433.33 nm
5. Ans. 9.3 μm
6. Ans. (i) 1 mm (ii) increase
7. Ans. $\frac{\lambda}{4\cos\alpha}$
8. Ans. 1.25 m
9. Ans. 7
10. Ans. 0, 125 nm
11. Ans. 5
12. Ans. 50